

Gr. XII RT – 09 VA (Date: 31.08.26)

Directions (1 to 7): Read the text given and answer the questions that follow.

Woodrow Wilson was referring to the liberal idea of the economic market when he said that the free enterprise system is the most efficient economic system. Maximum freedom means maximum productiveness; our “openness” is to be the measure of our stability. Fascination with this ideal has made Americans defy the “Old World” categories of settled possessiveness versus unsettling deprivation, the cupidity of retention versus the cupidity of seizure, a “status quo” defended or attacked. The United States, it was believed, had no status quo ante. Our only “station” was the turning of a stationary wheel, spinning faster and faster. We did not base our system on property but opportunity—which meant we based it not on stability but on mobility. The more things changed, that is, the more rapidly the wheel turned, the steadier we would be. The conventional picture of class politics is composed of the Haves, who want a stability to keep what they have, and the Have-Nots, who want a touch of instability and change in which to scramble for the things they have not. But Americans imagined a condition in which speculators, self-makers, runners are always using the new opportunities given by our land. These economic leaders (front-runners) would thus be mainly agents of change. The nonstarters were considered the ones who wanted stability, a strong referee to give them some position in the race, a regulative hand to calm manic speculation; an authority that can call things to a halt, begin things again from compensatorily staggered “starting lines.”

“Reform” in America has been sterile because it can imagine no change except through the extension of this metaphor of a race, wider inclusion of competitors, “a piece of the action,” as it were, for the disenfranchised. There is no attempt to call off the race. Since our only stability is change, America seems not to honor the quiet work that achieves social interdependence and stability. There is, in our legends, no heroism of the office clerk, no stable industrial work force of the people who actually make the system work. There is no pride in being an employee (Wilson asked for a return to the time when everyone was an employer). There has been no boasting about our social workers—they are merely signs of the system’s failure, of opportunity denied or not taken, of things to be eliminated. We have no pride in our growing interdependence, in the fact that our system can serve others, that we are able to help those in need; empty boasts from the past make us ashamed of our present achievements, make us try to forget or deny them, move away from them. There is no honor but in the Wonderland race we must all run, all trying to win, none winning in the end (for there is no end).

1. The primary purpose of the passage is to
 - (1*) criticize the inflexibility of American economic mythology
 - (2) contrast “Old World” and “New World” economic ideologies
 - (3) challenge the integrity of traditional political leaders
 - (4) champion those Americans whom the author deems to be neglected

2. According to the passage, “Old World” values were based on
 - (1) ability (2*) property
 - (3) family connections (4) guild hierarchies
3. In the context of the author’s discussion of regulating change, which of the following could be most probably regarded as a “strong referee” in the United States?
 - (1) A school principal (2) A political theorist
 - (3*) A federal court judge (4) A social worker
4. The author sets off the word “Reform” (line 35) with quotation marks in order to
 - (1) emphasize its departure from the concept of settled possessiveness
 - (2) show his support for a systematic program of change
 - (3) underscore the flexibility and even amorphousness of United States society
 - (4*) assert that reform in the United States has not been fundamental
5. Which of the following metaphors could the author most appropriately use to summarize his own assessment of the American economic system ?
 - (1) A windmill (2) A waterfall (3*) A treadmill (4) A gyroscope
6. It can be inferred from the passage that Woodrow Wilson’s ideas about the economic market
 - (1) encouraged those who “make the system work” (lines 45-46)
 - (2*) perpetuated traditional legends about America
 - (3) revealed the prejudices of a man born wealthy
 - (4) foreshadowed the stock market crash of 1929
7. The passage contains information that would answer which of the following questions?
 - I. What techniques have industrialists used to manipulate a free market?
 - II. In what ways are “New World” and “Old World” economic policies similar?
 - III. Has economic policy in the United States tended to reward independent action?
 - (1) I only (2) II only (3*) III only (4) I and II only
8. **Direction: Fill in the blank with the correct punctuation:**

The weather forecast predicted heavy rain ____ we decided to postpone the picnic.

 - (1*) ; (Semicolon) (2) , (Comma)
 - (3) ! (Exclamation mark) (4) ? (Question mark)
9. Pick the best substitute for the following expression:

A person who is unable to make mistakes or do anything wrong.

 - (1*) Infallible (2) Invincible (3) Irresistible (4) Indispensable
10. Pick the correct meaning of the phrase “BONA FIDE”:
 - (1*) In good faith; genuine (2) For the sake of argument
 - (3) After the event (4) By force of law
11. Pick the word with the correct spelling:
 - (1) Accomodation (2*) Accommodation (3) Acommodation (4) Accommadation

12. Pick the odd one out from the following options:

- (1) Pick up (2) Hand in (3) Call in (4*) Look into

13. Pick the active voice of the following sentence:

The injured player was being examined by the doctor.

- (1) The doctor examined the injured player.
(2) The doctor has examined the injured player.
(3*) The doctor was examining the injured player.
(4) The doctor had examined the injured player.

14. Pick the passive voice of the following sentence:

The police have arrested the suspect.

- (1) The suspect has arrested by the police.
(2) The suspect was arrested by the police.
(3*) The suspect has been arrested by the police.
(4) The suspect had been arrested by the police.

15. Pick the correct set of words to complete the following sentence:

The scientist's findings were met with considerable _____, as several experts questioned the _____ of his conclusions.

- (1*) skepticism, validity (2) admiration, uncertainty
(3) enthusiasm, weakness (4) approval, controversy

16. Pick the correct meaning of the underlined phrase in the following sentence:

The young entrepreneur is on the threshold of success.

- (1) He has already achieved great success.
(2*) He is very close to achieving success.
(3) He has given up his efforts to succeed.
(4) He is uncertain about whether to continue working.

17. Choose the word that is closest in meaning to the underlined word:

The chairman's response was evasive, leaving the journalists without a clear answer.

- (1) straightforward (2) deliberate
(3*) avoiding a direct answer (4) delusional

18. Pick the correct meaning of the underlined phrase:

The company decided to cut corners to complete the project before the deadline.

- (1) To find a shorter route to the destination
(2*) To reduce unnecessary expenditure or effort, often by compromising standards
(3) To divide the project into smaller sections
(4) To abandon a difficult project

19. Choose the option which conveys almost the same meaning as the underlined word:

Despite repeated warnings from the authorities, the recalcitrant employee refused to comply with the new regulations.

- (1) submissive (2*) obstinate (3) meticulous (4) apprehensive

20. Choose the option which best replaces the underlined phrase in more conventional English:

“The director behaved as if he had been aware of the problem all along, although the evidence suggests otherwise.”

- (1) as if he was aware of the problem all along
(2) as though he were aware of the problem all along
(3*) as if he had known about the problem all along
(4) as though he has known about the problem all along

Directions (21 to 23): Arrange the following sentences in the most logical order to form a coherent paragraph:

21.A. Yet this does not imply withdrawal from the world; rather, it changes the manner in which one participates in it.

B. Spiritual maturity is often mistaken for the absence of desire, conflict, or emotional disturbance.

C. The deeper transformation lies in recognising these experiences without allowing them to dictate one's sense of self.

D. A person may continue to experience ambition, disappointment, and attachment while becoming progressively less governed by them.

- (1) BCDA (2) CBAD (3*) BDCA (4) DBCA

22.A. Consequently, the goal of psychological well-being cannot simply be to eliminate every uncomfortable emotion.

B. Anxiety, grief, frustration, and loneliness are often treated as evidence that something has gone fundamentally wrong within us.

C. Such emotions can instead function as signals, drawing attention to needs, conflicts, losses, or situations that require adaptation.

D. What matters is whether an individual can acknowledge these states, interpret them appropriately, and respond without being overwhelmed by them.

- (1*) BACD (2) CBDA (3) BCAD (4) DCBA

- 23.A. The paradox is that the more urgently one tries to secure a permanent sense of certainty, the more vulnerable one becomes to the inevitable changes of experience.
- B. Philosophical reflection begins, therefore, not by supplying certainty but by questioning our assumption that certainty is necessary for a meaningful life.
- C. Human beings frequently seek certainty because uncertainty exposes the limits of their control over events, relationships, and even themselves.
- D. This vulnerability arises because what we regard as secure is continually dependent on circumstances that remain beyond our complete control.

(1) CDAB (2) BCAD (3) ACDB (4*) CADB

24. Which of the following statements are grammatically accurate regarding the use of “NONE”?

- A. “None” may function as a pronoun and can refer to both singular and plural antecedents.
- B. When “none” refers to a plural noun, a plural verb is always mandatory.
- C. In formal usage, “none of the information is reliable” is grammatically acceptable.
- D. “None” must invariably be followed by the preposition “of” when it functions as a pronoun.

(1*) A and C Only (2) B and D Only (3) A, B and C Only (4) A, C and D Only

25. **Direction: Match LIST-I with LIST-II according to the grammatical function of the underlined word.**

LIST-I		LIST-II	
A.	The committee has <u>still</u> not reached a consensus.	I.	Adjective
B.	The <u>then</u> minister rejected the proposal.	II.	Adverb
C.	The proposal was <u>beyond</u> the scope of the inquiry.	III.	Noun
D.	The committee examined the <u>outside</u> before approving the design.	IV.	Preposition

(1) A-II, B-I, C-III, D-IV

(2) A-I, B-II, C-III, D-IV

(3*) A-II, B-I, C-IV, D-III

(4) A-III, B-I, C-II, D-IV

Directions (26 to 32): Read the given text and answer the questions that follow.

Ten per cent of life is made up of what happens to you. Ninety per cent of life is decided by how you react. Most of the unpleasant incidents in life can be averted if we choose not to react. We often react because we mistakenly think that if we don't, the other person will assume we're afraid. It's our fragile ego that makes us react violently. A reaction gives birth to yet another reaction, and this goes on. There's no end to it. Reaction is a sudden manifestation of anger. It's a sign of impulsive behaviour. So, not reacting is always sagacious because non-reaction frustrates the other person. In other words, non-reaction is the best action.

When an angry villager abused the Buddha, he coolly asked the villager, "Son, do you still have anything to say?" That man was utterly ashamed. We can all follow this because reaction intensifies action, making it difficult to control. When fire breaks out in the jungle, it's drawn to dry leaves, and from there, it spreads all over the forest. Reactions are like dry leaves; they attract actions to turn into conflagration. Avoid them if you want to live peacefully.

Behavioural psychologists are of the view that to react is human. It's in our DNA. So, what should we do? Someone very wittily said that the human itch to react can be replaced with responding. A response is always a watered-down reaction. The difference between responding and reacting is a choice. You must decide whether to react or respond. By reacting rather than responding, we indirectly reveal our level of emotional intelligence. At the same time, a reaction is a telltale of our maturity (or lack thereof) and inner fabric. To quote Urdu poet Hasrat Nazim, " – my reply revealed my character, had I chosen to keep mum, I'd have been called an intelligent person. Not everything and everyone needs your reaction. Reacting is recklessness. Protect your peace and keep walking.

Sometimes, the most powerful response involves observing, reflecting, and choosing silence or inaction. Just because you don't react doesn't mean you don't notice. Silence isn't always a sign of ignorance; it can be the strongest form of observation. You'll continue to suffer if you've an emotional reaction to everything said to you. True power is sitting back and observing everything with logic. True power is restraint. If words control you that means everything else can control you. Breathe and allow things to pass. A Chinese proverb states that when you don't react, you irritate the person who expects a reaction from you. He may lose his peace of mind, but you retain your calm.

Non-reaction is extreme indifference, even apathy; it's your non-verbal reaction which involves ignoring all that's petty. Provocation needn't necessarily be returned with provocation. At the moment, we're living in a reactionary world with reactionary world leaders, whose impulsive reactions are creating endless miseries.

26. Which of the following most accurately captures the **central distinction** developed throughout the passage between reacting and responding?

- (1*) Reacting is an involuntary emotional impulse, whereas responding involves consciously regulating the impulse before acting.
- (2) Reacting is an involuntary emotional impulse, whereas responding involves consciously suppressing the impulse before acting.
- (3) Reacting is an instinctive emotional response, whereas responding involves deliberately eliminating the emotion before acting.
- (4) Reacting is an instinctive emotional response, whereas responding involves deliberately expressing the emotion before acting.

27. The statement “**non-reaction frustrates the other person**” most strongly implies which of the following?
- (1) Provocation often depends partly on the expectation that the target will emotionally participate in the conflict.
 - (2) Provocation often succeeds only when the target consciously decides to emotionally participate in the conflict.
 - (3*) Provocation loses much of its intended force when the target refuses to emotionally participate in the conflict.
 - (4) Provocation loses all of its intended force when the target consciously refuses to emotionally participate in the conflict.
28. The image of “dry leaves” in the passage primarily symbolises:
- (1*) Emotional vulnerabilities that allow an initial disturbance to spread into progressively larger conflicts.
 - (2) Emotional impulses that allow an initial disturbance to develop into progressively larger conflicts.
 - (3) Emotional vulnerabilities that allow an isolated disturbance to disappear into progressively larger conflicts.
 - (4) Emotional impulses that allow an isolated disturbance to disappear into progressively larger conflicts.
29. The author's recommendation that one should “**respond**” rather than “**react**” rests most fundamentally on which assumption?
- (1*) Individuals generally possess sufficient control over their actions to interrupt an impulsive emotional sequence.
 - (2) Individuals generally possess sufficient control over their emotions to prevent an impulsive emotional sequence.
 - (3) Individuals generally possess sufficient awareness of their emotions to eliminate an impulsive emotional sequence.
 - (4) Individuals generally possess sufficient control over circumstances to interrupt an impulsive emotional sequence.
30. Which of the following would the author of the passage be **LEAST likely to endorse**?
- (1) Remaining silent during provocation may sometimes demonstrate greater maturity than offering an immediate verbal defence.
 - (2*) Remaining silent during provocation may sometimes demonstrate greater maturity than suppressing an immediate emotional impulse.
 - (3) Remaining silent during provocation may sometimes demonstrate greater wisdom than offering an immediate emotional defence.
 - (4) Remaining silent during provocation may sometimes demonstrate greater restraint than offering an immediate verbal defence.

31. Which statement, if true, would **most seriously weaken** the author's claim that non-reaction is generally the wiser course?
- (1) In some conflicts, refusing to react may encourage the aggressor to interpret silence as permission and intensify the provocation.
 - (2) In some conflicts, refusing to react may encourage the aggressor to interpret silence as fear and intensify the provocation.
 - (3*) In some conflicts, refusing to react may encourage the aggressor to interpret silence as weakness and intensify the provocation.
 - (4) In some conflicts, refusing to react may encourage the aggressor to interpret silence as indifference and intensify the provocation.
32. The passage's concluding assertion that “**non-reaction is extreme indifference, even apathy**” appears, at first glance, to conflict with its earlier advocacy of non-reaction. Which interpretation best resolves this apparent contradiction?
- (1) The author distinguishes deliberate non-reaction as restraint from indiscriminate non-reaction as indifference.
 - (2*) The author distinguishes deliberate non-reaction as restraint from habitual non-reaction as indifference.
 - (3) The author distinguishes emotional non-reaction as restraint from habitual non-reaction as indifference.
 - (4) The author distinguishes deliberate non-reaction as restraint from emotional non-reaction as indifference.

Direction (33 and 34): Choose the most appropriate option reporting the statement given.

33. The philosopher said, “If you had not allowed your anger to control you yesterday, you would not be regretting your words today.”
- (1) The philosopher said that if I had not allowed my anger to control me yesterday, I would not have regretted my words that day.
 - (2) The philosopher said that if I had not allowed my anger to control me the previous day, I would not have been regretting my words that day.
 - (3) The philosopher said that if I did not allow my anger to control me the previous day, I would not be regretting my words that day.
 - (4*) The philosopher said that if I had not allowed my anger to control me the previous day, I would not be regretting my words that day.
34. The researcher asked the students, “Why did you assume that the theory was false merely because you could not immediately understand it?”
- (1*) The researcher asked the students why they had assumed that the theory had been false merely because they could not immediately understand that.
 - (2) The researcher asked the students why had they assumed that the theory was false merely because they could not immediately understand it.
 - (3) The researcher asked the students why they had assumed that the theory was false merely because they could not immediately understand it.
 - (4) The researcher asked the students why they assumed that the theory had been false merely because they had not immediately understood it.

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35. In an election 10% candidates do not vote and $\frac{100}{9}\%$ votes are declared invalid. Winning candidate gets 75% of the valid votes. If he is more than 2000 votes then what is the number of registered votes?
- (1) 50000 (2) 20000 (3) 15000 (4) None of these
36. Due to an international supply-chain disruption, the wholesale price of a specialty manufacturing polymer increased by 32%. A manufacturing firm wishes to restrict the increase in its total expenditure on this polymer to only 10%. By what percentage must the firm reduce its raw polymer procurement volume so as to meet its target budget?
- (1) 14.50% (2*) $16\frac{2}{3}\%$ (3) 18.25% (4) 20.00%
37. A wholesale commodity dealer purchases basmati rice from farmers using a tampered balance that reads 900 grams when it actually receives 1000 grams of rice. He then marks up the price by 40% above the true cost price. While selling to retail consumers, he uses another rigged scale that reads 1000 grams while dispensing only 800 grams of rice. Furthermore, he offers an attractive trade discount of 10% to cash-paying customers on the marked price. What is the dealer's actual overall profit percentage earned on the entire transaction?
- (1*) 75% (2) 70% (3) 68% (4) 80%
38. A retailer purchased 500 electronic tablets at an identical cost price. He sold 40% of the consignment at a profit of 25%, 30% of the consignment at a profit of 15%, and 20% of the consignment at a discount of 5% on the cost price. Due to transit damage, the remaining tablets were sold as refurbished units at a loss of 35% on the cost price. If his overall net profit across the entire batch of 500 tablets was ₹49,000, find the cost price of a single tablet.
- (1) ₹920 (2) ₹950 (3*) ₹980 (4) ₹1,020
39. Mihir's capital is $\frac{5}{4}$ times more than Tulsi's capital. Tulsi invested her capital at 50% per annum for 3 years (compounded annually). At what rate % p.a. simple interest should Mihir invest his capital so that after 3 years, they both have the same amount of capital?
- (1) $\frac{20}{3}\%$ (2) 10% (3) $\frac{50}{3}\%$ (4) None of these
40. A businessman borrows ₹25,220 from a financial institution at an interest rate of 5% per annum, compounded annually. The debt is to be completely cleared in three equal annual installments paid at the end of each year. Find the exact numerical value of each installment.
- (1) ₹8,820 (2*) ₹9,261 (3) ₹9,500 (4) ₹9,800

41. Three identical casks A , B , and C of volume 60 liters each contain mixtures of ethanol and water in the volume ratios $3 : 1$, $4 : 1$, and $5 : 1$ respectively. Exactly 16 liters from Cask A , 20 liters from Cask B , and 24 liters from Cask C are drawn and poured into an empty fourth vessel D . What is the ratio of ethanol to water in vessel D ?
- (1*) $4 : 1$ (2) $7 : 2$ (3) $9 : 2$ (4) $5 : 1$
42. The monthly incomes of three professionals X , Y and Z are in the ratio $4 : 5 : 6$, and their monthly expenditures are in the ratio $3 : 4 : 5$. If X saves exactly 25% of his monthly income, find the ratio of the monthly savings of X , Y and Z .
- (1*) $1 : 1 : 1$ (2) $2 : 3 : 4$ (3) $3 : 4 : 5$ (4) $4 : 5 : 6$
43. A batch of 120 engineering students is divided into three sections P , Q and R containing 35, 45, and 40 students respectively. In a national aptitude assessment, the average score of Section P was 72, Section Q was 78, and the average score of all three sections combined was 75. If two highest scorers with scores of 98 and 92 from Section R are excluded, what is the new average score of the remaining students in Section R ?
- (1*) 73.16 (2) 74.50 (3) 72.80 (4) 75.25
44. A fast bowler in a premier cricket league has a bowling average of 16.8 runs per wicket. In the championship final, he bowls a spell of 6 wickets for 33 runs, which brings down his career bowling average by exactly 0.6 runs per wicket. What was the total number of wickets taken by the bowler in his entire career up to the conclusion of the final match?
- (1) 107 (2*) 113 (3) 118 (4) 124
45. A civil engineering contractor undertakes to construct a bridge in 150 days and employs 80 workers. After 90 days, he discovers that only $\frac{1}{2}$ of the total construction work has been accomplished. How many additional workers of equal efficiency must he deploy immediately to complete the bridge strictly on schedule?
- (1) 30 (2*) 40 (3) 50 (4) 60
46. Two inlet pipes A and B can fill an empty reservoir in 15 hours and 20 hours respectively, while a bottom drainage pipe C can completely empty the full reservoir in 30 hours. All three pipes are opened together at 6:00 AM. Pipe A is closed at 10:00 AM, and Pipe C is closed at 1:00 PM. At what time will the reservoir be completely full?
- (1) 11:40 PM (2) 12:20 AM (3*) 1:20 AM (4) 2:00 AM
47. If 3 men, 4 women, and 6 boys can complete a manufacturing assignment in 12 days working together. A man works twice as fast as a boy, and a woman works 1.5 times as fast as a boy. In how many days can 4 men, 6 women, and 7 boys complete a work that is thrice the size of the original assignment?
- (1) 24 days (2*) 27 days (3) 30 days (4) 32 days
48. Two express trains T_1 and T_2 start simultaneously from New Delhi and Mumbai respectively towards each other. After passing each other, T_1 takes 9 hours to reach Mumbai and T_2 takes 16 hours to reach New Delhi. If the speed of T_1 is 80 km/h, find the total distance between New Delhi and Mumbai.
- (1) 1,560 km (2*) 1,680 km (3) 1,720 km (4) 1,800 km

49. Two athletes A and B run on a circular track of circumference 600 m starting from the same point at the same time in the same direction. The speed of A is 15 m/s and the speed of B is 9 m/s. After how much time will A meet B for the third time at the starting point?
- (1) 6 minutes (2) 8 minutes (3*) 10 minutes (4) 12 minutes
50. A motorboat can travel 48 km upstream and 72 km downstream in a total of 9 hours. In the same river, it can travel 72 km upstream and 48 km downstream in a total of 11 hours. Find the speed of the river current in km/h.
- (1*) 8 km/h (2) 6 km/h (3) 4 km/h (4) 10 km/h
51. The product of the HCF and LCM of two numbers is 2160. If the HCF of the numbers is 6, both numbers are two-digit integers, and the difference between the two numbers is 42, find the sum of the two numbers.
- (1*) 102 (2) 114 (3) 120 (4) 96
52. If the nine-digit number $785x3678y$ is completely divisible by 72, find the value of the algebraic expression $(5x - 3y)$.
- (1) 14 (2*) 18 (3) 22 (4) 26
53. Find the unit digit of the expression: $E = 1! + 2! + 3! + \dots + 100! + 3^{2026} \times 7^{2026}$.
- (1) 2 (2*) 4 (3) 6 (4) 8
54. Let $N = 2^5 \times 3^4 \times 5^3 \times 11^2$. How many factors of N are even, divisible by 15, and perfect squares?
- (1) 6 (2*) 8 (3) 12 (4) 16
55. Find the remainder when 2^{1000} is divided by 33.
- (1*) 1 (2) 2 (3) 31 (4) 32
56. Let α and β be the roots of the quadratic equation $x^2 - 5x + 3 = 0$. Find the value of $\alpha^4 + \beta^4$.
- (1) 327 (2*) 343 (3) 355 (4) 361
57. For what real value of k does the following system of linear equations have infinitely many solutions?
- $$2x + 3y + 4z = 9$$
- $$4x + ky + 8z = 18$$
- $$6x + 9y + 12z = 27$$
- (1) 3 (2*) 6 (3) 8 (4) 12
58. In an Arithmetic Progression (AP), the ratio of the sum of the first m terms to the sum of the first n terms is $\frac{m^2}{n^2}$. Find the ratio of the m^{th} term to the n^{th} term.
- (1) $\frac{m}{n}$ (2*) $\frac{2m-1}{2n-1}$ (3) $\frac{2m+1}{2n+1}$ (4) $\frac{m-1}{n-1}$
59. The sum of the first three terms of a Geometric Progression is 13 and their product is 27. If the common ratio $r > 1$, find the 5th term of the GP.
- (1) 27 (*2) 81 (3) 243 (4) 729
60. If $x = 7 + 4\sqrt{3}$, evaluate the numerical value of $\sqrt{x} + \frac{1}{\sqrt{x}}$.
- (1) $2\sqrt{3}$ (2*) 4 (3) $4\sqrt{3}$ (4) 8

61. Solve for x in the logarithmic equation: $\log_3 (x) + \log_9 (x) + \log_{27} (x) = 11$.
 (1) 243 (2*) 729 (3) 2187 (4) 81
62. In $\triangle ABC$, the side lengths are $AB = 8$ cm, $AC = 12$ cm, and $BC = 10$ cm. The internal angle bisector of $\angle A$ meets side BC at point D . Find the length of the segment BD .
 (1*) 4 cm (2) 5 cm (3) 6 cm (4) 4.5 cm
63. A solid metallic right circular cone of base radius 12 cm and height 18 cm is melted and recast into a solid right circular cylinder of height 6 cm. Find the curved surface area of the resulting cylinder (in cm^2).
 (1) $120\pi \text{ cm}^2$ (2*) $144\pi \text{ cm}^2$ (3) $160\pi \text{ cm}^2$ (4) $180\pi \text{ cm}^2$
64. A cylindrical container of internal diameter 28 cm is partially filled with water. Exactly 21 solid spherical lead balls of diameter 4 cm each are completely submerged into the water. By how much (in cm) will the water level rise in the container?
 (1*) $\frac{8}{7}$ cm (2) $\frac{6}{7}$ cm (3) 1.5 cm (4) $\frac{9}{7}$ cm
65. From an external point P , a tangent PT of length 12 cm is drawn to a circle. A secant PAB passes through the center of the circle, intersecting the circumference at A and B . If $PA = 8$ cm, find the radius of the circle.
 (1) 4 cm (2*) 5 cm (3) 6 cm (4) 7.5 cm
66. In an academic batch of 100 students, 65 study Mathematics, 45 study Physics, and 40 study Chemistry. Exactly 25 study both Mathematics and Physics, 20 study both Physics and Chemistry, and 15 study both Mathematics and Chemistry. If 5 students study none of the three subjects, how many students study all three subjects?
 (1) 3 (2) 5 (3) 8 (4) 10
67. In how many ways can 6 male executives and 6 female executives be seated around a circular conference table such that no two female executives sit adjacent to each other?
 (1*) 86,400 (2) 43,200 (3) 14,400 (4) 518,400

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68.If 26 January, 2011 was Wednesday, then what day of the week was it on 26th January, 2020?

- (1) Monday (2) Tuesday (3) Friday (4*) Sunday

69.If it was Saturday on December 17, 1899, then what will be the day on December 22, 1905?

- (1) Monday (2*) Thursday (3) Friday (4) Sunday

70.On which dates of April, 2012 will a Sunday come?

- (1) 5, 12, 19, 26 (2*) 1, 8, 15, 22, 29 (3) 3, 10, 17, 24 (4) 7, 14, 21, 28

Directions (71 to 73): Read the following information carefully and answer the questions given below.

The outer border of width 1 cm of a cube with side 5 cm is painted yellow on each side and the remaining space enclosed by this 1 cm path is painted pink. This cube is now cut into 125 smaller cubes of each side 1 cm. The smaller cubes so obtained are now separated.

71.How many smaller cubes have all the surfaces uncoloured?

- (1) 0 (2) 9 (3) 18 (4*) 27

72.How many smaller cubes have three surfaces coloured yellow?

- (1) 2 (2) 4 (3*) 8 (4) 10

73.How many smaller cubes have one face coloured pink and an adjacent face yellow?

- (1*) 0 (2) 1 (3) 2 (4) 4

Direction (74 and 75): In each of the following questions, there is a certain relationship between two given words on one side of (: :) and one word is given on another side (: :) while another word is to be found from the given alternatives, having the same relation with this word as the words of the given pair bear. Choose the correct alternative.

74.Lion : Roar : : Ass : ?

- (1) Bark (2) Trumpet (3) Howl (4*) Bray

75.63 : 21 : : 27 : ?

- (1) 6 (2*) 9 (3) 1 (4) 3

76.If in a certain code language 'CHOLINE' is written as 'OCIHLEN', then how will 'SURGEON' be written in that language?

- (1) RSUEGNO (2) RSEGUNO (3) RESUGNO (4*) RSEUGNO

77.If 'MUSICAL' is written as 'KWQKACJ', then how can 'SPRINKLE' be written?

- (1) QRBKCNJG (2) QNPGLIJC (3*) QRPKLMJG (4) URTKPMNG

78. There are 25 boys in a horizontal row. Rahul was shifted by three places towards his right side and he occupies the middle position in the row. What was his original position from the left end of the row?

- (1) 15th (2) 16th (3) 12th (4*) 10th

79. In a row of girls, Veena is 12th from the start and 19th from the end. In another row of girls, Sunita is 14th from the start and 20th from the end. How many girls are there in both the rows together?

- (1) 72 (2) 65 (3*) 63 (4) 61

Direction (80 to 81): In the following series, replace the question mark (?) with the suitable option.

80. 360, 180, 90, 45, 22.5, ?

- (1) 15 (2*) 11.25 (3) 20 (4) 10.5

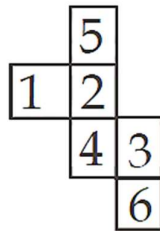
81. 0, 4, 18, 48, ?, 180

- (1) 58 (2) 68 (3) 84 (4*) 100

82. Find out the odd numbers/number pairs from the given alternatives.

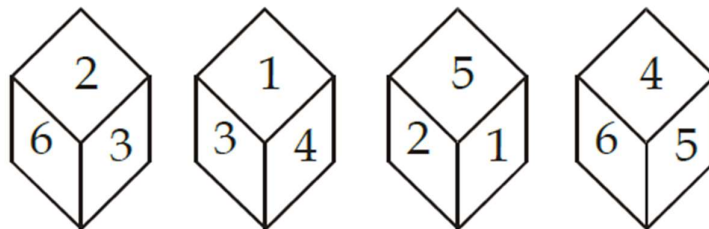
- (1) 31 (2) 13 (3*) 49 (4) 19

83. When the following figure is folded to form a cube, then which number is at the opposite of number 1?



- (1*) 3 (2) 6 (3) 4 (4) 5

84. **Direction:** In the following question, four positions of the same dice has been shown. You have to see these figure and select the number/symbol opposite to number/symbol as asked in the question.



Which number is at the opposite of number 3?

- (1*) 5 (2) 1 (3) 2 (4) 6

85. Radha moves towards South-East a distance of 7 m, then she moves towards West and travels a distance of 14 m. From here, she moves towards North-west a distance of 7 m and finally she moves a distance of 4 m towards East and stood at that point. How far is the starting point from where she stood?

- (1) 3 m (2) 4 m (3*) 10 m (4) 11 m

86. I am facing South. I turn right and walk 20 m. Then I turn right again and walk 10 m. Then I turn left and walk 10 m and then turning right walk 20 m. Then I turn right again and walk 60 m. In which direction am I from the starting point?

- (1) North (2) North-West (3) East (4*) North-East

Directions (87 to 89): Read the information given below and then answer the questions that follow:

Eight friends M, N, O, P, Q, R, S and T are sitting in a circle facing the centre. P is third to the right of M and second to the left of S. T is third to the left of S. Q is not immediate neighbour of S. N is not immediate neighbour of M. O is second to the right of N. Q is also not immediate neighbour of T.

87. Who is third to the right of N?

- (1*) Q (2) T (3) P (4) R

88. Who is second to the left of M?

- (1) N (2) R (3) S (4*) None of these

89. Who is sitting between P and S?

- (1) Only R (2) R and N (3*) Only N (4) None of these

90. A clock is set right at 12:00 noon. The clock loses 20 minutes in 24 hours. How many minute the clock will lose till 6 am next day?

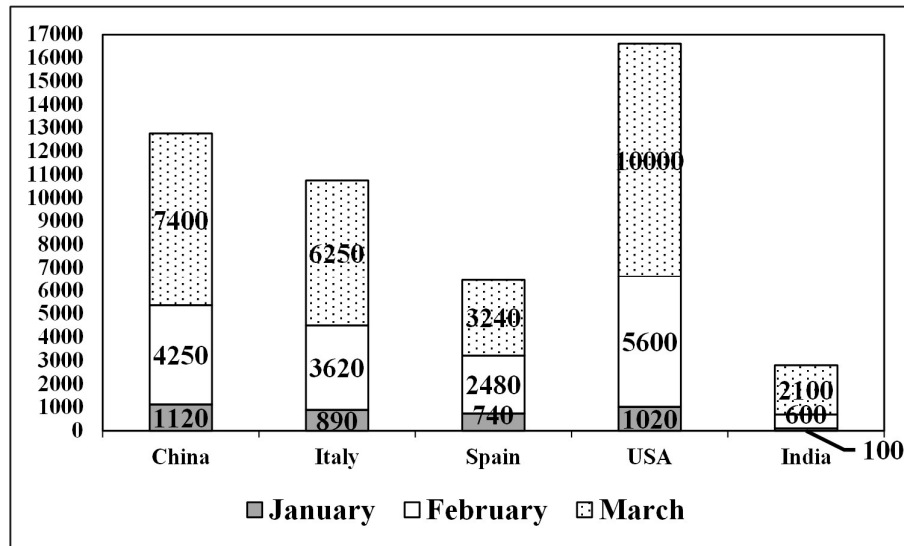
- (1) 12 min (2) 20 min (3) 18 min (4*) 15 min

Gr. XII RT – 09 DI (Date: 31.08.26)

Directions (91 to 95): Study the following bar chart carefully and answer the questions given beside.

Data regarding the number of people who were tested positive with COVID-19 during January, February and March in five countries is given in the stack bar chart.

Data of number of People in five countries who were found positive with COVID-19 test



91. Find the average number of people per day who were tested positive in India in March.

- (1) 60.73 (2*) 67.74 (3) 72.34 (4) 76.47

92. Which countries in February showed more than 3000 positive tests?

- (1) China and Italy only (2) China and the USA only
(3) Italy and the USA only (4*) China, Italy and the USA only

93. By what percent the number of positive tested people grew in Spain in February from January?

- (1) 335.13% (2*) 235.13% (3) 353.13% (4) 253.13%

94. Japan in January had twice the number of cases that India had in January while 50% more cases in February than India that India had in February. Find the number of cases in Japan in March if cases in March were twice the total cases till February end.

- (1) 900 (2) 1000 (3*) 2200 (4) 2000

95. Find the ratio of the number of cases in February and March together in USA to the number of cases in China in February and the number of cases in Italy in March together.

- (1) 41: 52 (2) 55 : 47 (3) 54 : 35 (4*) None of these

Directions (96 to 100): Study the following table chart carefully and answer the questions given beside.

Two movies Bahubali and Robot 2.0 are released in three theatres INOX, PVR and Galaxy. On Friday a total of 4500 people watched Bahubali and 4000 people watched Robot 2.0.

The table below shows the percent distribution of total viewers of a particular movie into three theatres.

Theatre	Bahubali	Robot 2.0
INOX	40%	35%
PVR	35%	25%
Galaxy	25%	40%

The viewers are classified on the basis of age as adults and children. The table below shows the percentage of children out of the total viewers in each theatre for each movie.

Theatre % of Children	Bahubali	Robot 2.0
INOX	15%	20%
PVR	16%	18%
Galaxy	20%	17%

96. What is the ratio of the number of children who watched Bahubali in PVR to the children who watched Robot in PVR?

- (1) 9 : 11 (2) 7 : 13 (3) 9 : 7 (4*) 7 : 5

97. The number of children who watched Bahubali in INOX is what percent of the number of adults who watched Bahubali in Galaxy?

- (1) 32% (2*) 30% (3) 45% (4) 27%

98. What is the difference between the total viewers in INOX and PVR?

- (1*) 625 (2) 650 (3) 480 (4) 520

99. What is the ratio of the sum of adults and children who watched Robot in INOX and PVR respectively to the sum of adults and children who watched Bahubali in Galaxy and INOX respectively?

- (1) 14 : 11 (2) 11 : 9 (3*) 10 : 9 (4) 13 : 7

100. The average number of children who watched Robot is what percent of the average number of adults who watched Bahubali?

- (1) 21.33% (2) 15.78% (3) 18.67% (4*) 19.50%

ANSWER KEY

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	3	4	3	2	3	1	1	1	2	4	3	3	1
Q. No.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	2	2	3	3	1	4	1	3	1	3	1	1	2
Q. No.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	2	4	1		2	1	3		2	1	1	1	2	2
Q. No.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	2	2	3	1	1	2	2	2	1	2	2	2	2	2
Q. No.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	2	1	2	1	2	2	1	4	2	2	4	3	1	4	2
Q. No.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	4	3	4	3	2	4	3	1	1	3	4	1	4	3	4
Q. No.	91	92	93	94	95	96	97	98	99	100					
Ans.	2	4	2	3	4	4	2	1	3	4					